

Walkthrough Agent

Teach a product walkthrough once in plain English. The agent performs it live in a real browser at call time.

Interactive demo tools capture an application and replay it. The captured artifact is unbreakable precisely because it is no longer the product — it drifts the moment the real app changes, and it produces nothing. This agent stores **intent, not pixels**. At call time it drives a real browser session against the real site, narrating each step from what the page actually did. The demo can fail, and that is the point: a run that completes is evidence the workflow still works.

HOW A WORKFLOW IS CREATED

1 • Compile — app/teach/compiler.py

Input is one English sentence plus an entry URL — no click recording. An LLM emits a draft of 3–6 steps, each an action plus an intent. **No locators exist yet**; nothing has touched a page. Safety is enforced in code rather than prompt text: destructive intents (delete, purchase, transfer, merge) hard-block the draft, a submit step is permitted only if the utterance declared a test/dummy target, and off-domain drift is rejected.

2 • Rehearse — app/teach/rehearser.py

A real Chrome session is attached over CDP and the draft is actually executed. Per step: snapshot the page, resolve the plain-language intent against the **live accessibility tree**, act, snapshot again, and capture the observed effect as step knowledge. Locators are discovered from what is on screen, never invented. An unresolvable step aborts rehearsal and raises a clarifying question — no partial workflow is ever written.

3 • Freeze — app/store.py

Only a fully rehearsed spec is persisted, written atomically to `WA_WORKFLOWS_DIR/{id}.v{n}.json`. Nothing becomes executable until it has survived one real run.

DESIGN DECISIONS WORTH DEFENDING

- **Ranked locator candidates.** Each step carries ordered fallback strategies (role, then text, then label). When the first stops matching, the driver tries the next before failing — the answer to page drift without re-teaching.
- **Safety checked twice.** Risk is assigned per step at compile time and re-checked by the execution guard at run time, so a stored workflow cannot be replayed into something destructive.
- **Narration derived from observed state.** Step text is assembled after the action resolves, from the live URL and matched element — not read from a prepared script.
- **Human interruption mid-run.** A chat channel runs alongside execution; the operator can pause the walkthrough between steps and ask questions answered from captured step knowledge.
- **Split-plane delivery.** FastAPI + WebSocket serves the operator UI over a public tunnel while execution stays on the driving machine. The interface is remote; the browser provably is not.

STATUS

Stack	Python, FastAPI, WebSocket, Playwright over CDP, Gemini
Tests	63 passing
Working	Teach pipeline, live execution, narration, interruption, tunnelled public UI
Config	Single <code>.env</code> ; no secrets in the repo

WHAT HAPPENS AT CALL TIME

The orchestrator loads the frozen spec and drives the same real Chrome session step by step. Each step resolves its locator against the page as it exists now, acts, and emits narration built from the resulting state; the safety guard re-checks step risk before anything is performed. The operator UI — a glowing step list, live narration, and a chat box — is served over the tunnel to whoever is watching, and a pause request lands between steps rather than mid-action, so a paused run never leaves a half-filled form. A step that cannot resolve surfaces the mismatch out loud instead of silently skipping, which is the one behaviour a recorded walkthrough can never exhibit.

KNOWN LIMITS

Execution requires the agent process on the driving machine — a browser page cannot script a third-party origin, so a URL alone can never move the work to the viewer's device. A packaged binary or browser extension is the route there, deliberately out of scope for this build. Recovery is limited to locator fallback; a genuine layout change still requires re-teaching, which is a one-sentence operation by design. The approach fits demos that must produce a real artifact — a record actually created — rather than a convincing picture of one.

Teaching input, verbatim: *“On github.com/<user>/<test-repo>, open the issues page, click New issue, fill in a title and description, and submit it. This is a dummy test repo.”* — ten rehearsed steps, persisted as a frozen spec, replayed live on demand.